LENSES

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

(A) An object is placed 20 cm away from a concave lens with a focal length of 10 cm.

(a) Calculate the distance to the image from the lens.

(b) Write down the characteristics of the image.

OR

(B) A convex lens forms a virtual image at a distance of 7.5 cm away from it. Focal length of the lens is 15 cm.

(a) Calculate the distance between lens and the object.

(b) Write any two instances where a convex lens is used in this manner.

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2. 2026 **2 MARKS**

Draw a ray diagram showing the formation of a virtual image by a convex lens.

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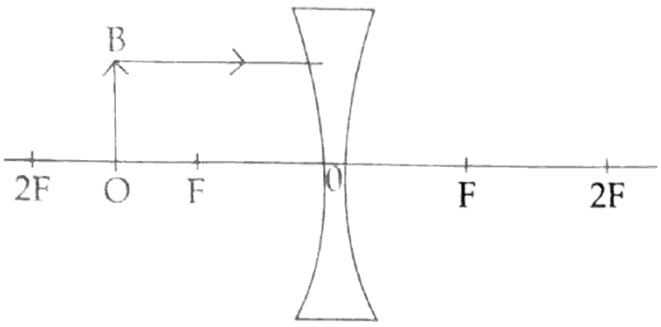
3. 2025 **4 MARKS**

Ray diagram of an object placed on the principal axis of a concave lens is shown below.

(a) Copy the diagram and complete it to show the image formation.

(b) The image formed by a concave lens is always virtual. Why ?

(c) What are the characteristics of the image formed here ?



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4. 2025 **2 MARKS**

Power of lens is -0.5 D.

(a) What is the relation between the power of a lens and its focal length ?

(b) Calculate the focal length of this lens.

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5. 2025 **1 MARK**

The midpoint of a lens is called _____ .

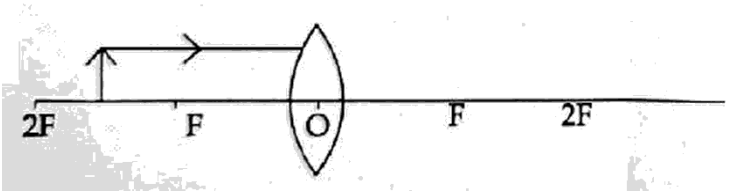
(Centre of curvature, Optic centre, Principal focus, Aperture)

6. 2024 **3 MARKS**

Observe the diagram of an object placed on the principal axis of a convex lens.

(a) Copy down the ray diagram and complete it to show the position of the image.

(b) Write the characteristics of the image obtained.



7. 2024 **1 MARK**

The power of a lens is +2D. Find its focal length.

8. 2023 **4 MARKS**

An object is placed at a distance of 15 cm from a lens of focal length +10 cm.

(a) Which type of lens is used here ?

(b) Calculate the image distance.

(c) What is the nature of the image formed ?

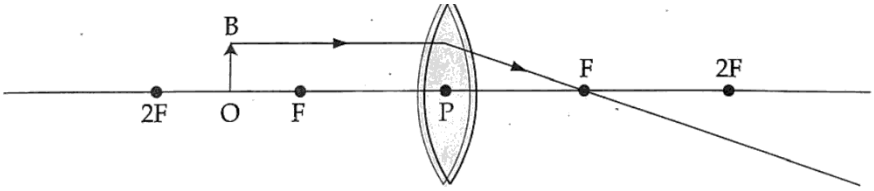
9. 2022 **4 MARKS**

Analyse the figure. An object is placed between F and 2F of a convex lens.

(a) Copy the diagram and complete to show the image formation.

(b) Write any two features of the image formed here.

(c) Where must the object be placed to get a real image of same size as that of the object.



10. 2022 **1 MARK**

The midpoint of a lens is known as _____ .

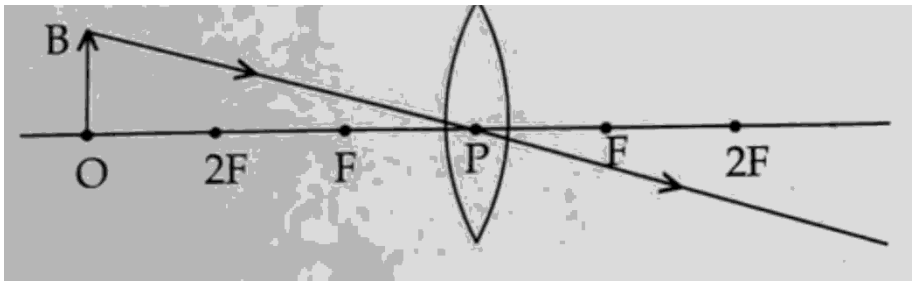
(Optic centre, Principal focus, Centre of curvature, Principal axis)

11. 2021 **4 MARKS**

Observe the ray diagram given below.

(a) Redraw the diagram and complete it to get the image.

(b) Write any two characteristics of the image obtained.



12. 2021 **4 MARKS**

(a) If the image obtained from a convex lens is erect and enlarged :

(i) Image is formed at : [Same side of the object / Opposite side of the object]

(ii) Write any one application of this type of image formation.

(b) Write any two characteristics of the image formed by a convex lens when the object is placed at the following positions.

(i) At infinity

(ii) Between F and 2F

13. 2021 **1 MARK**

The image formed by a convex lens is inverted and diminished. Then the object must be placed :

[At 2F, Beyond 2F, Between F and 2F, At F]